Upsampling

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Key Concepts

1) Upsampling increases the sampling rate by an integer factor M.

    a) If the original sampling interval is T, the up sampled sampling interval is T/M.

    b) If the original sampling frequency Ω_s, the up sampled sampling frequency is $M\Omega_s$.

2) Upsampling is performed in practice by first inserting $M-1$ zeros in between every sample and then applying a discrete-time lowpass filter with stop band $|\omega| > \frac{\pi}{M}$.

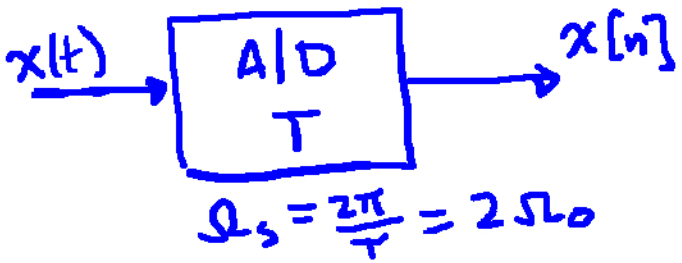
a) Inserting $M-1$ zeros scales the discrete-time Fourier transform frequency (DTFT) axis by M and maps the interval $-\pi < \omega \leq \pi$ into the interval $-\frac{\pi}{M} < \omega \leq \frac{\pi}{M}$.

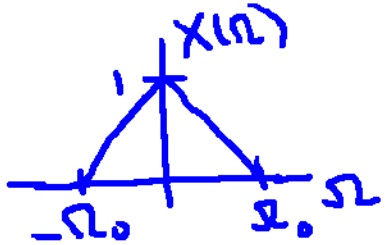
b) The lowpass filter eliminates the replicates of the signal DTFT in $\frac{\pi}{M} < |\omega| \leq 2\pi - \frac{\pi}{M}$ that are introduced by the frequency axis scaling.

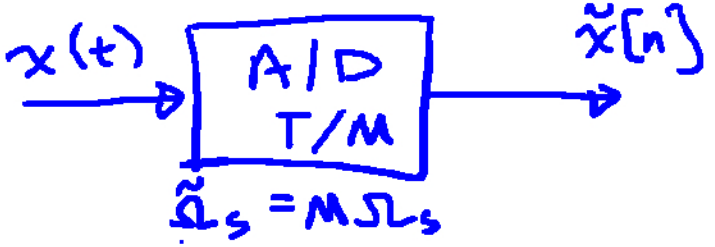
c) Viewed in the time domain, the lowpass filter smooths the sharp transitions caused by the introduction of zeros between samples.

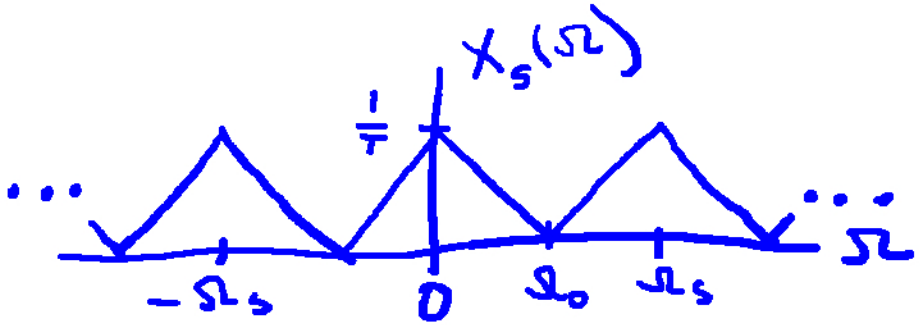
Video Screenshots

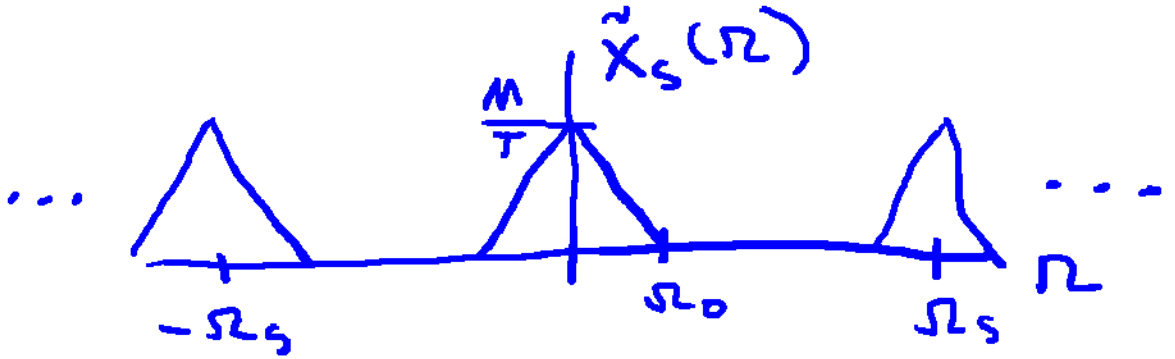
Upsampling (Discrete-time Interpolation)



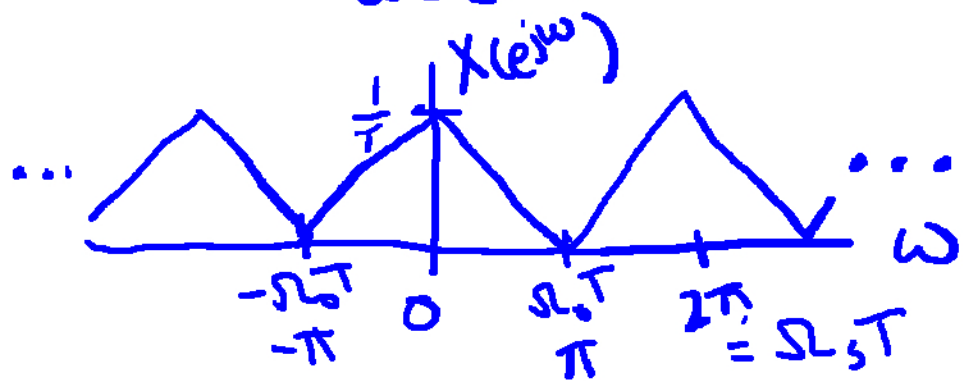


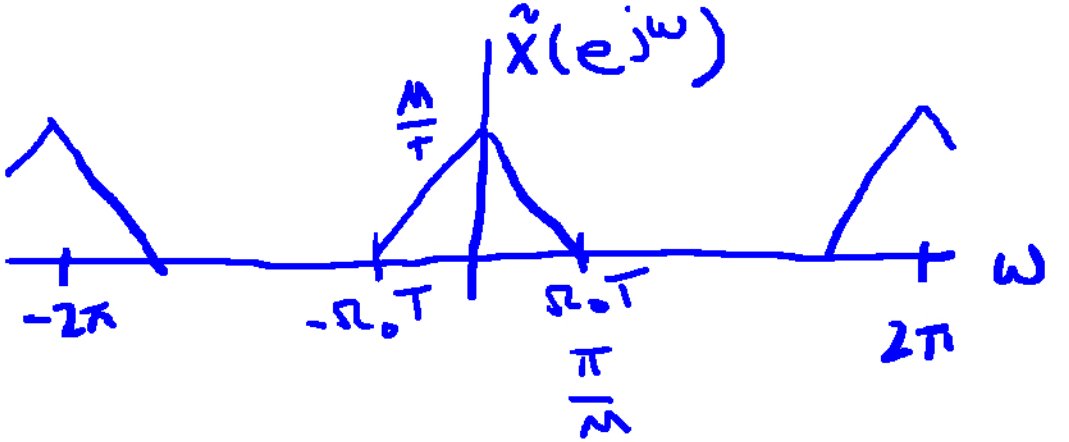


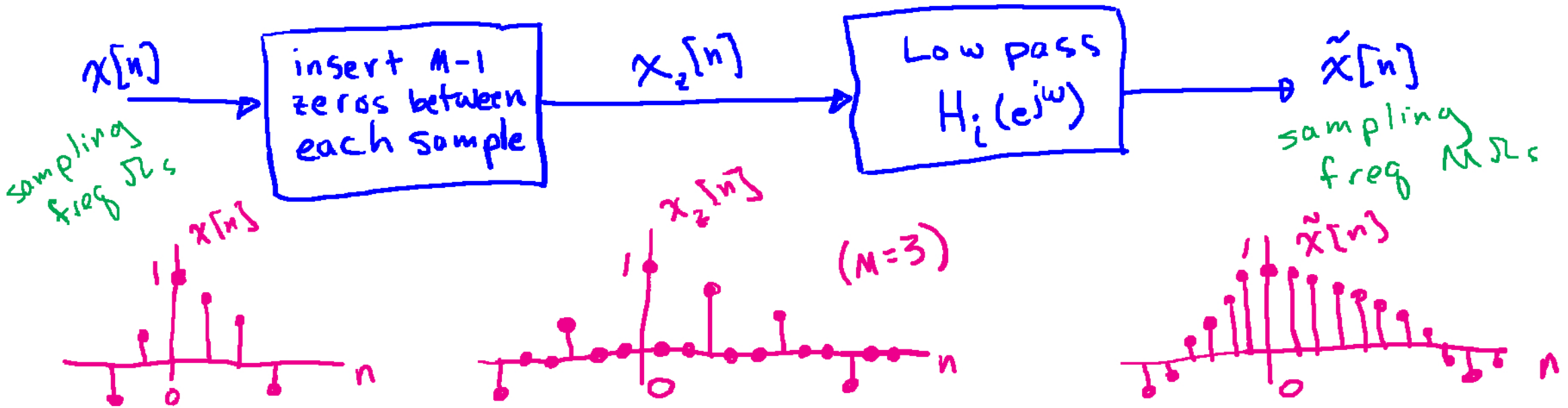




$\omega = \Omega T$







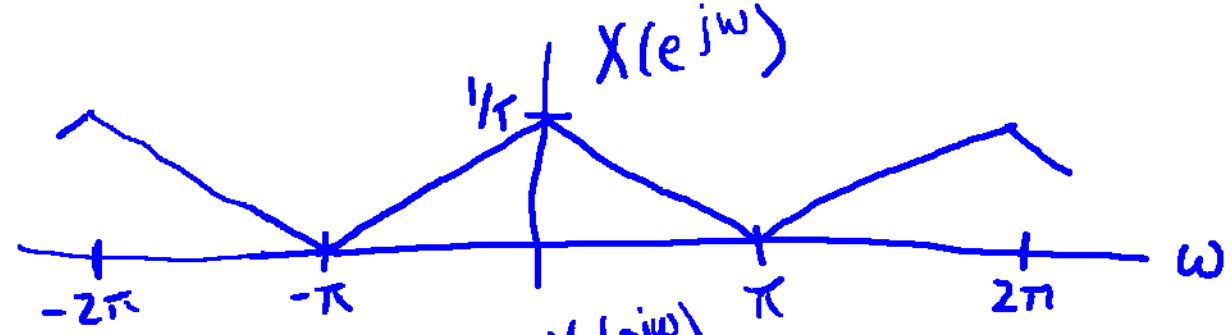
$$x_z[n] = \begin{cases} x[n/M], & n/M \text{ integer} \\ 0, & \text{other} \end{cases} \quad \text{or} \quad x_z[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-kM]$$

So

$$X_z(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} x[k]\,\delta[n-kM] \right) e^{-j\omega n} = \sum_{k=-\infty}^{\infty} x[k] \left(\sum_{n=-\infty}^{\infty} \delta[n-kM] e^{-j\omega n} \right)$$

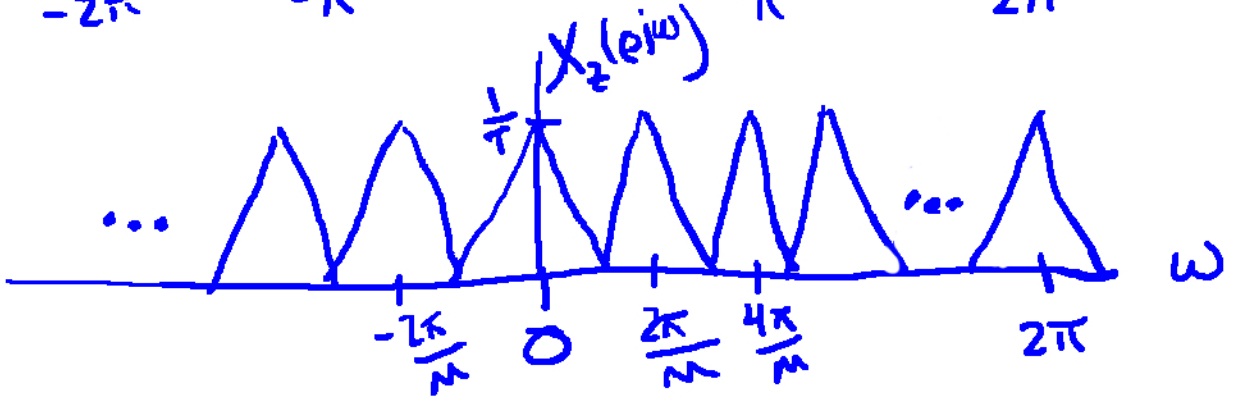
$$e^{-j\omega Mk}$$

$$= \sum_{k=-\infty}^{\infty} x[k]\, e^{-j \underbrace{\omega M}_{\omega'} k} = X(e^{j\omega M}) \quad \text{scale } \omega \text{ by } M$$

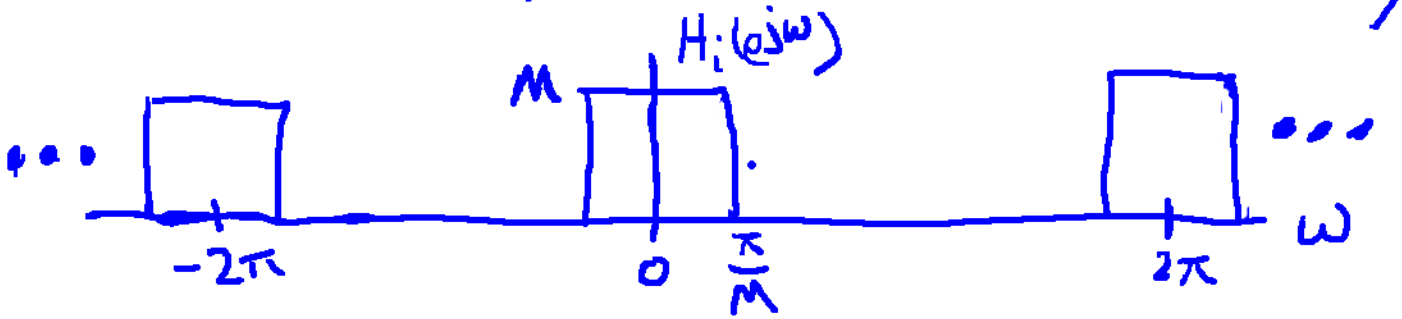


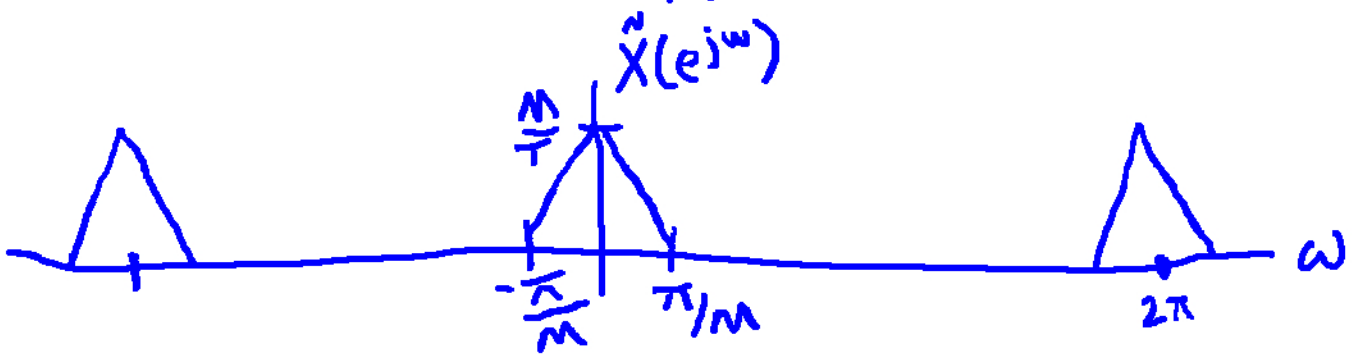
scale axis by M

$X_z(e^{j\omega}) = X(e^{j\omega M})$



$\times$ filter by $H_i(e^{j\omega})$





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